

Samuel K. Grunblatt

CONTACT	335 Gallalee Hall, Dept. of Physics & Astronomy	sgrunblatt@ua.edu
INFORMATION &	University of Alabama	skgrunblatt.github.io
CITIZENSHIP	ORCID # 0000-0003-4976-9980	Citizen of U.S.A. and Germany
EMPLOYMENT	University of Alabama , Tuscaloosa, AL Assistant Professor, Department of Physics and Astronomy, 2024– Johns Hopkins University , Baltimore, MD Assistant Research Scientist, Department of Physics and Astronomy, 2022–2024 American Museum of Natural History/Flatiron Institute , New York, NY Research Associate, Department of Astrophysics, 2022–2024 Kalbfleisch Fellow, Department of Astrophysics, 2019–2022 Guest Researcher, Center for Computational Astrophysics, 2019–2023	
EDUCATION	University of Hawaii–Manoa , Honolulu, HI Ph.D., Astronomy, August 2019 M.S., Astronomy, Dec 2015 Columbia University , New York, NY B.A., Astrophysics, May 2013	
GRANTS, AWARDS AND TELESCOPE TIME	Identifying Movement-Based Biomarkers..., <i>NSF</i> SBIR, \$300,000, Co-I Planetary architectures of evolved..., <i>TESS</i> Space Telescope, \$70,000, PI Constraining heating mechanisms of evolved..., WIYN/NEID, 5n, PI Testing (re)inflation of planets..., TNG/HARPS, 3n, Science PI Measuring the Mass of a Benchmark Hot Jupiter, Magellan/PFS, 1n, PI Checkmate: atmospheres of planets., VLT/ESPRESSO, 1.6n, Co-I Investigating Planet (Re-)Inflation..., CTIO/CHIRON, 4n, PI TESS’s Ear on the Metal-Poor Milky Way, \$70,000, Co-I Planetary Archaeology:..., NASA-Keck I/HIRES, 6n, \$70,512, PI Planetary Archaeology:..., <i>TESS</i> Space Telescope, \$70,000, PI Constraining Orbits of Hot Jupiters Around Evolved Stars, TESS DDT, PI Planetary Archaeology:..., <i>TESS</i> Space Telescope, \$50,000, PI Measuring Long Rotation..., <i>TESS</i> Space Telescope, \$200,000, Co-I RV Follow-up..., SALT/HRS, 2n, PI Planetary Archaeology:..., <i>TESS</i> Space Telescope, \$50,000, Science PI UH–Manoa Student Excellence in Research Award, Honolulu, HI	Jul 2025 Sep 2024 2024B– 2024A– 2024A 2023B 2023A–2024A May 2022 2021A–2023A May 2021 Feb 2021 Jun 2020 Jun 2020 Nov 2019– Jun 2019 Apr 2018
PUBLICATION SUMMARY	40 publications in peer-reviewed journals (10 1st-author, 7 2nd-author) 450+ first-author citations, 1050+ total citations, h-index=19, 1st-author h-index=9 My full bibliography is available on NASA ADS .	
RECENT INVITED CONFERENCE TALKS AND SEMINARS	Dynamical Synthesis of Planetary Systems, Yale University Physics & Astronomy Dept. Colloquium, Western Kentucky University Astronomy Dept. Colloquium, University of Florida TESS Extended Mission Splinter Session, AAS #245 Astronomy Dept. Colloquium, University of Massachusetts–Amherst Physics and Astronomy Dept. Colloquium, University of Alabama Astronomy Colloquium, Rochester Institute of Technology Exoplanet Lunch, Princeton University Exoplanet Seminar, Ohio State University	Jan 2026 Oct 2025 Sep 2025 Jan 2025 Nov 2024 Feb 2024 Nov 2023 Oct 2023 Sep 2023

	European Astronomical Society Annual Meeting, Krakow, Poland	July 2023
	Roman Science Inspired by Emerging JWST Results Meeting, STScI	June 2023
	Astrophysics Seminar, George Washington University	May 2023
	Exoplanet Seminar, NASA Goddard Space Flight Center	Apr 2023
	SES Brownbag Seminar, JHU Applied Physics Laboratory	Apr 2023
SELECTED PRESS	“Mysterious ‘Phoenix’ World Challenges Theories...,” Newsweek	Jun 2024
	“Spunky Exoplanet Inexplicably Survives...,” Gizmodo	Jun 2024
	“This Fiery Jupiter-Sized World...,” Inverse	Jan 2022
	“When Giant Planes Orbit Evolved Stars,” AAS Nova	Jul 2018
	“How Old Stars Make Hot Jupiter Exoplanets So Huge,” IB Times	Nov 2017
	“Puffed-up exoplanets inflate with heat from their stars alone,” New Scientist	Sep 2016
TEACHING AND MENTORING EXPERIENCE	AY203: <i>Observational Astronomy</i> , U. Alabama	Fall 2024–
	Guest Lecturer, <i>Planets, Life, and the Universe</i> , Johns Hopkins U.	Sep 2023
	Guest Lecturer, <i>Language of Astrophysics</i> , Johns Hopkins U.	Nov 2022
	Guest Lecturer & Course Designer, <i>Astronomy Lab for Majors</i> , U. Hawaii	2016-2018
Postgraduate		
	Ph.D. Committee Member for N. Derksen, U. Alabama	June 2025–
	Ph.D. Committee Member for B. Akintunde, U. Alabama	June 2025–
	Ph.D. Committee Member for I. Ethige, U. Alabama	Apr 2025–
	Advisor to S. Balaji, PhD, U. Alabama	Apr 2025–
	Advisor to E. Page, PhD, Lehigh University (now Alabama)	Aug 2022–
	Advisor to N. Saunders, PhD, UH–Manoa (now Yale)	Sep 2019–June 2025
Undergraduate		
	Advisor to B. Nnadi, JHU Rowland Summer Research Fellow	2023
	Advisor to N. Sodickson, ISEF/JSJS (now UChicago)	Jun 2023–
	Advisor to K. Gary, AMNH REU (now OSU, NSF Honorable Mention)	Jun 2020–
	Advisor to S. Yoshida, ISEF/JSJS (now Harvard)	Jun 2017–Jan 2023
PROFESSIONAL SERVICE AND OUTREACH	Science Organizing Committee, NASA Sagan Summer Workshop	2025–2026
	Grant Reviewer for NASA, NSF, Swiss NSF	2023–
	TESS Science Conference II Splinter Session Lead Organizer	Aug 2021
	Seminar Organizing Committee, American Museum of Natural History	2019–2022
	Lead Co-Organizer, CCA Stars and Exoplanets Meeting	2019–2022
	Time Allocation Committee, HST, NOIRLab, U. Hawaii	2017–
	Journal Referee for <i>A&A</i> , <i>AJ</i> , <i>ApJ</i> , <i>ApJL</i> , <i>ApJS</i> , <i>PASJ</i> , <i>MNRAS</i>	2016–
	American Astronomical Society Member	2015–
REFERENCES	Daniel Huber	
	Associate Professor	Phone: 808-956-8573
	Institute for Astronomy	E-mail: huberd@hawaii.edu
	University of Hawaii	
	Ruth Angus	
	Associate Curator (faculty)	Phone: 212-313-3581
	Department of Astrophysics	E-mail: rangus@amnh.org
	American Museum of Natural History	
	Kevin Schlaufman	
	Associate Professor	Phone: 410-516-3295
	Department of Physics and Astronomy	E-mail: kschlaufman@jhu.edu
	Johns Hopkins University	

1. **Grunblatt, S.**, Saunders, N., Huber, D., et al. “Giants Transiting Giants IX: Seven additional hot Jupiters around evolved stars found by TESS,” *AAS Journals*, *in prep.*
2. † Page, E., **Grunblatt, S.**, Saunders, N., Pepper, J. “Giants Transiting Giants VIII: Applying Machine Learning Classification Tools to Identify New Planet Candidates Transiting Evolved Stars,” *to be submitted.*
3. † Sodickson, N., & **Grunblatt, S.**, “In Search of Decay: An Updated Analysis of the Transit Times of Hot Jupiters,” *ApJ*, *accepted.*
4. Hammerberg, A., **Grunblatt, S.**, & Kramer, P. “Movement Sequencing: A Novel Approach to Quantifying the Building Blocks of Human Gait,”
<https://arxiv.org/pdf/2402.07893>
5. † Saunders, N., **Grunblatt, S.**, Page, E., Huber, D., Chontos, A., et al. “Giants Transiting Giants VII: A Hot Saturn Orbiting an Oscillating Red Giant Star,” *AJ*, 169, 75, 2025.
6. **Grunblatt, S.** “Giant branch systems: surveys and populations,” *Encyclopedia of Astrophysics*, 1st ed., *in press*, <https://arxiv.org/pdf/2410.16482>
7. † Saunders, N., **Grunblatt, S.**, Dai, F., Chontos, A., Huber, D., et al. “TESS Giants Transiting Giants VI. Newly Discovered Hot Jupiters Provide Evidence for Efficient Obliquity Damping After the Main Sequence,” *AJ*, 168, 81, 2024. [8 citations]
8. **Grunblatt, S.**, Saunders, N., Huber D., Yoshida, S., Vissapragada, S., et al. “TESS Giants Transiting Giants IV. An unlikely survivor: a low-density hot Neptune orbiting a red giant star,” *AJ*, 168, 1, 2024. [9 citations]
9. Eisner, N., **Grunblatt, S.**, Barragan, O., Blunt, S., Saunders, N., et al. “A bright, nearby, multiplanet binary star system with a transiting gas planet in the habitable zone,” *AJ*, 167, 241, 2024. [6 citations]
10. † Nnadi, B., & **Grunblatt, S.**, “Galactic Archaeology with Luminous Red Giant Oscillations in Gaia DR3 Photometry,” *RNAAS*, 8, 59, 2024.
11. † Pereira, F., **Grunblatt, S.**, Psaridi, A., Saunders, N., Campante, T., et al. “TESS Giants Transiting Giants V. Two hot Jupiter systems around evolved stars in the southern ecliptic hemisphere,” *MNRAS*, 527, 6332, 2024. [7 citations]
12. **Grunblatt, S.**, Wilson, R., Winter, A., Gaudi, B., Huber, D., et al. “Adding Fields Hosting Globular Clusters To The Galactic Bulge Time Domain Survey,” Roman Core Community Survey White Paper, arXiv:2306.10647. [5 citations]
13. **Grunblatt, S.**, Saunders, N., Chontos, A., Hattori, S., Veras, D., et al. “TESS Giants Transiting Giants III. An eccentric warm Jupiter supports a period-eccentricity relation for giant planets transiting evolved stars,” *AJ*, 165, 44, 2023. [13 citations]
14. † Yoshida, S., **Grunblatt, S.**, & Price-Whelan, A. “Determining the Detectability of Planets Transiting Stars of Extragalactic Origin,” *AJ*, 164, 119, 2022. [1 citation]

15. **Grunblatt, S.**, Saunders, N., Sun, M., Chontos, A., Soares-Furtado, M., et al. “TESS Giants Transiting Giants II. The hottest Jupiters orbiting evolved stars,” *AJ*, 163, 120, 2022. [33 citations]
16. † Saunders, N., **Grunblatt, S.**, Huber, D., Collins, K., Brahm, R. et al. “TESS Giants Transiting Giants I. A Non-inflated Hot Jupiter Orbiting a Massive Subgiant,” *AJ*, 163, 53, 2022. [22 citations]
17. **Grunblatt, S.**, Zinn, J., Price-Whelan, A., Angus, R., Saunders, N., et al. “Age-Dating Red Giant Stars Associated with Galactic Disk and Halo Substructures,” *ApJ*, 916, 88, 2021. [32 citations]
18. † Yoshida, S., **Grunblatt, S.**, Hermes, J., Armstrong, J., Coughlin, J. et al. “Eclipsing Binary and White Dwarf Features Associated with K2 Target EPIC251248385,” *RNAAS*, 3, 174, 2019.
19. **Grunblatt, S.**, Huber, D., Gaidos, E., Hon, M., Zinn, J., et al. “Giant planet occurrence within 0.2 AU of low-luminosity red giant branch stars with K2,” *AJ*, 158, 227, 2019. [48 citations]
20. **Grunblatt, S.**, Huber, D., Gaidos, E., Lopez, E., Barclay, T., et al. “Do close-in giant planets orbiting evolved stars prefer eccentric orbits?,” *ApJL*, 861, L5, 2018. [35 citations]
21. **Grunblatt, S.**, Huber, D., Gaidos, E., Lopez, E., Howard, A., et al. “Seeing double with K2: Testing re-inflation with two remarkably similar planets around red giant branch stars,” *AJ*, 154, 254, 2017. [77 citations]
22. **Grunblatt, S.**, Huber, D., Gaidos, E., Lopez, E., Fulton, B., et al. “K2-97b: A (Re-?)Inflated Planet Orbiting a Red Giant Star,” *AJ*, 152, 185, 2016. [73 citations]
23. **Grunblatt, S.**, Howard, A., & Haywood, R. “Determining the Mass of Kepler-78b With Nonparametric Gaussian Process Estimation,” *ApJ*, 808, 127, 2015. [140 citations]

CO-AUTHORED
JOURNAL
PUBLICATIONS

1. Lindsay, C., Ong, J., Basu, S., **Grunblatt, S.**, Hon, M. “The Effect of Different Methods for Accounting for α -enhancement on the Asteroseismic Modeling of Metal-Poor Stars,” *AAS Journals*, *in review*.
2. Tang, J., Wang, S., Li, Y., Bedding, T., Xiao, G., et al. including **Grunblatt, S.** “RV×TESS I: Modeling Asteroseismic Signals with Simultaneous Photometry and RVs,” *AAS Journals*, *in review*.
3. Chontos, A., Huber, D., **Grunblatt, S.**, Saunders, N., Winn, J., et al. “The TESS-Keck Survey XXI: 13 New Planets and Homogeneous Properties for 21 Subgiant Systems,” *AAS Journals*, *in review*, <https://arxiv.org/pdf/2402.07893> [10 citations]
4. Howard, A., Sinukoff, E., Blunt, S., Petigura, E., Crossfield, I., et al. including **Grunblatt, S.** “Planet Masses, Radii, and Orbits from NASA’s K2 Mission,” *ApJS*, 278, 52, 2025. [1 citation]
5. Huber, D., Slumstrup, D., Hon, M., Li, Y., Børsen-Koch, V., et al. including **Grunblatt, S.** “Stellar Models are Reliable at Low Metallicity: An Asteroseismic Age for the Ancient Very Metal-Poor Star KIC 8144907,” *ApJ*, 975, 19, 2024. [11 citations]

6. Schmidt, S., Schlaufman, K., Ding, K., **Grunblatt, S.**, Carmichael, T., et al. “Verification of Gaia’s Single-lined Spectroscopic Binary Solutions With Three Low-mass Secondaries,” *AJ*, 166, 225, 2023. [7 citations]
7. Mallorquin Diaz, M., Goffo, E., Palle, E., Lodieu, N., Bejar, V., et al. including **Grunblatt, S.** “TOI-1801 b: a temperate mini-Neptune around a young M0.5 dwarf,” *A&A*, 680, 76, 2023. [9 citations]
8. Huber, D., Pinsonneault, M., Beck, P., Bedding, T., Bland-Hawthorn, J., et al. including **Grunblatt, S.** “Asteroseismology with the Roman Galactic Bulge Time-Domain Survey,” Roman Core Community Survey White Paper, arXiv:2307.03237. [15 citations]
9. Blunt, S., Carvalho, A., David, T., Beichman, C., Zink, J., et al. including **Grunblatt, S.** “Overfitting Affects the Reliability of Radial Velocity Mass Estimates of the V1298 Tau Planets,” *AJ*, 166, 62, 2023. [35 citations]
10. Lin, Z., Gan, T., Wang, S., Shporer, A., Rabus, M., et al. including **Grunblatt, S.** “Three low-mass companions around aged stars discovered by TESS,” *MNRAS*, 523, 6162, 2023. [9 citations]
11. Knudstrup, E., Gandolfi, D., Nowak, G., Persson, C., Furlan, E., et al. including **Grunblatt, S.** “Radial velocity confirmation of a hot super-Neptune with a warm Saturn-mass companion discovered by TESS,” *MNRAS*, 519, 5637, 2023. [3 citations]
12. Vissapragada, S., Chontos, A., Greklek-Mckee, M., Knutson, H., Dai, F., et al. including **Grunblatt, S.** “The Possible Tidal Demise of Kepler’s First Planetary System,” *ApJL*, 941, 31, 2022. [37 citations]
13. Tayar, J., Moyano, F., Soares-Furtado, M., Escorza, A., Joyce, M., et al. including **Grunblatt, S.** “Spinning up the Surface: Evidence for Planetary Engulfment or Unexpected Angular Momentum Transport?” *ApJ*, 940, 23, 2022. [13 citations]
14. Stello, D., Saunders, N., **Grunblatt, S.**, Hon, M., Reyes, C., et al. “TESS asteroseismology of the Kepler red giants,” *MNRAS*, 476, 536, 2022. [45 citations]
15. Gaidos, E., Hirano, T., Kraus, A.L., Kuzuhara, M., Zhang, Z., et al. including **Grunblatt, S.** “Zodiacal Exoplanets in Time (ZEIT) XII: A Directly-Imaged Planetary-Mass Companion to a Young Taurus M Dwarf Star,” *MNRAS*, 3069, 2819, 2021. [10 citations]
16. Murphy, J., Kosiarek, M., Batalha, N., Gonzales, E., Isaacson, H., including **Grunblatt, S.** “Another super-dense sub-Neptune in K2-182 b and refined mass measurements for K2-199 b and c,” *AJ*, 162, 294, 2021. [13 citations]
17. Polanski, A., Crossfield, I., Burt, J., Nowak, G., Lopez-Morales, M., including **Grunblatt, S.** “Wolf 503 b: Characterization of a Sub-Neptune Orbiting a Metal-Poor K Dwarf,” *AJ*, 162, 238, 2021. [10 citations]
18. Zhang, J., Weiss, L., Huber, D., Blunt, S., Chontos, A., et al. including **Grunblatt, S.** “A Long-Period Giant Planet Causes Inner Planets to be Misaligned with the Host Star in Kepler-129 System,” *AJ*, 162, 89, 2021. [23 citations]
19. David, T., Contardo, G., Sandoval, A., Angus, R., Lu, Y., et al. including **Grunblatt, S.** “Evolution of the Exoplanet Size Distribution: Forming Super-Earths Over Billions of Years,” *AJ*, 161, 265, 2021. [49 citations]

20. Kosiarek, M., Berardo, D., Crossfield, I., Laguna, C., Akana Murphy, J., et al. including **Grunblatt, S.** “Physical Parameters of the Multi-Planet Systems HD 106315 and GJ 9827,” *AJ*, 161,47, 2021. [18 citations]
21. Lu, Y., Angus, R., Agueros, M., Blancato, K., Ness, M., et al. including **Grunblatt, S.** “Astraea: Predicting Long Rotation Periods with 27 Day Light Curves,” *AJ*, 160, 168, 2020. [20 citations]
22. Huber, D., Chaplin, W. J.; Chontos, A., Kjeldsen, H., Christensen-Dalsgaard, J., et al. including **Grunblatt, S.** “A hot Saturn orbiting an oscillating late subgiant discovered by TESS,” *AJ*, 157, 245, 2019. [96 citations]
23. White, T. R., Huber, D., Mann, A. W., Casagrande, L., **Grunblatt, S. K.**, et al. “Interferometric diameters of five evolved intermediate-mass planet-hosting stars measured with PAVO at the CHARA Array,” *MNRAS*, 447, 4, 2018. [48 citations]
24. North, T., Campante, T., Miglio, A., Davies, G., **Grunblatt, S.**, et al. “Weighing in on the masses of retired A stars with asteroseismology. Kepler and K2 observations of exoplanet hosts,” *MNRAS*, 472, 1866, 2017. [29 citations]
25. Fulton, J., Howard, A., Weiss, L., Sinukoff, E., Petigura, E., et al. including **Grunblatt, S.** “Three Temperate Neptunes Orbiting Nearby Stars,” *ApJ*, 830, 46, 2016. [29 citations]
26. Younes, G., Kouveliotou, C., van der Horst, A.J., Baring, M. G., Granot, J. et al. including **Grunblatt, S.** “Time resolved spectroscopy of SGR J1550-5418 bursts detected with Fermi/GBM,” *ApJ*, 785, 52, 2014. [32 citations]